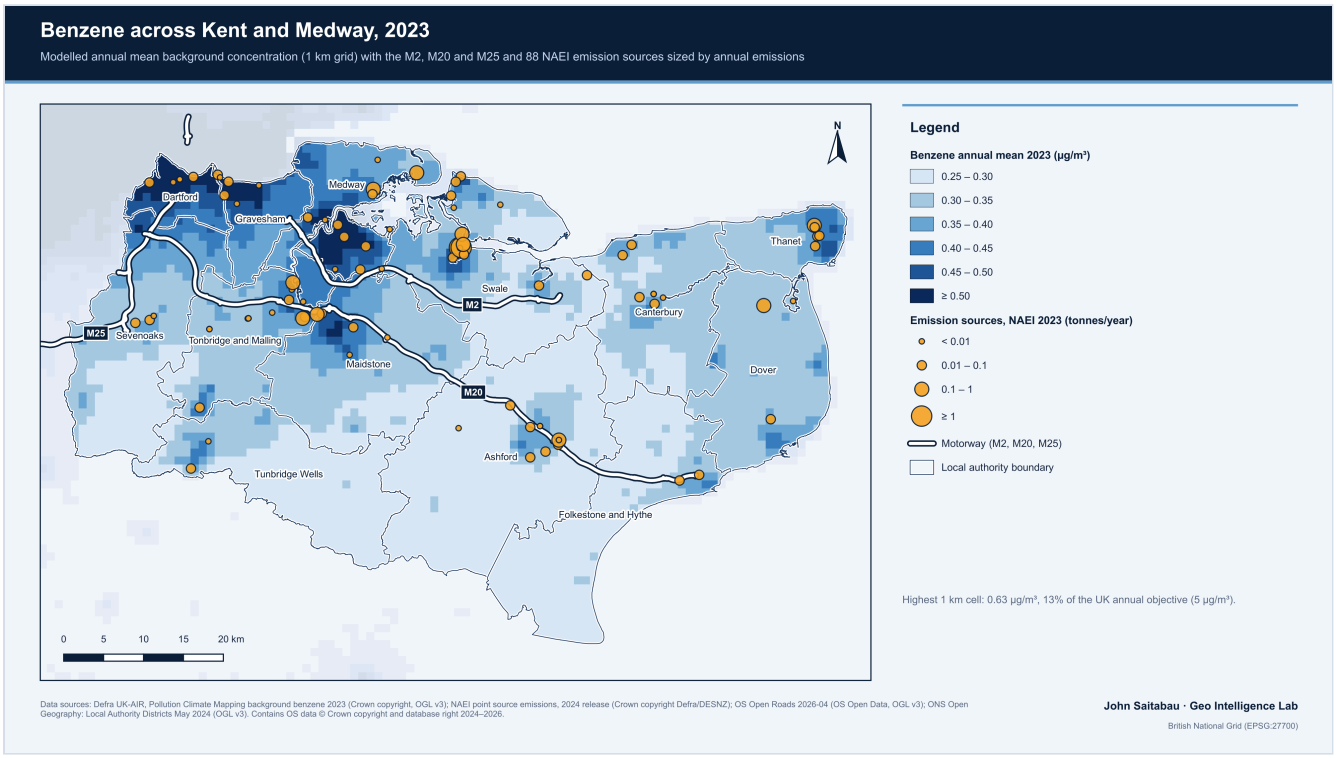
GIS PORTFOLIO CASE STUDY

Background benzene and the people who live with it

Kent and Medway, 2023: a reproducible exposure analysis from open UK data



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Tools
QGIS 3.34, PyQGIS, Python

Study area
12 Kent districts and Medway

Date
October 2026

At a glance

Benzene causes leukaemia, and the WHO sets no level below which it is considered harmless. This study asks two things: **where in Kent and Medway is background benzene highest**, and **how many residents live in those places?**

0.63

µg/m³ peak 1 km cell

13% of the UK annual objective and 37% of the WHO reference level

337,566

residents at or above 0.50 µg/m³

18.2% of the population; 92% of them in three north-west authorities

ρ = 0.11

emissions vs local concentration

88 point sources released 12.3 t, but barely shape the map

The short answer

Every part of the county sits far below the legal limit. The modelled annual mean runs from 0.25 to 0.63 µg/m³. The pattern is still uneven. Concentrations rise towards the Thames estuary and its towns, and Dartford averages 1.8 times the level of Ashford. The residents in the top band are clustered in Medway, Dartford and Gravesham. The large industrial emitters, such as the paper-mill power plants at Kemsley, do not line up with the highest cells. That points to traffic and the urban background, not single plants, as the main driver.

What this project shows

- **Data engineering:** five public sources downloaded by script and logged with URL, date, SHA-256 and licence
- **Raster analysis:** converting a national CSV grid to a raster, clipping it, and area-weighted zonal statistics
- **Population exposure:** Census 2021 residents allocated through 5,734 Output Area centroids
- **Spatial statistics:** rank correlation between emitters and concentration, and threshold sensitivity
- **Cartography:** three print layouts in British National Grid with a shared class scheme
- **Reproducibility:** seven numbered PyQGIS and Python scripts rebuild every number and map from scratch

Finding 1 · Levels are low everywhere

No 1 km cell comes near a standard (see the overview map on the cover). The highest, in Dartford, reaches 0.63 µg/m³: about one-eighth of the UK objective of 5 µg/m³ and just over a third of the WHO reference level of 1.7 µg/m³. The population-weighted mean for all 1,855,811 residents is 0.416 µg/m³. So the question is not whether Kent breaches a limit. It is where the relative burden falls, and on whom.

Data and method

Data sources

Source	What was used	Licence
Defra UK-AIR, Pollution Climate Mapping	Modelled background benzene, annual mean 2023, 1 km grid (µg/m³)	OGL v3.0
National Atmospheric Emissions Inventory	Benzene from point sources, 2023 (2024 release), tonnes/year	Crown copyright Defra/DESNZ
ONS Open Geography Portal	Local Authority Districts (May 2024); Output Area population-weighted centroids (2021)	OGL v3.0
ONS Census 2021 via Nomis	TS001 usual residents by Output Area	OGL v3.0
OS Open Roads	Motorways (M2, M20, M25), for map context	OGL v3.0

All layers are in British National Grid (EPSG:27700). Every download is logged in `raw/MANIFEST.csv` with its URL, retrieval date and SHA-256 checksum. Contains public sector information licensed under the Open Government Licence v3.0. Contains OS data © Crown copyright and database right 2024–2026.

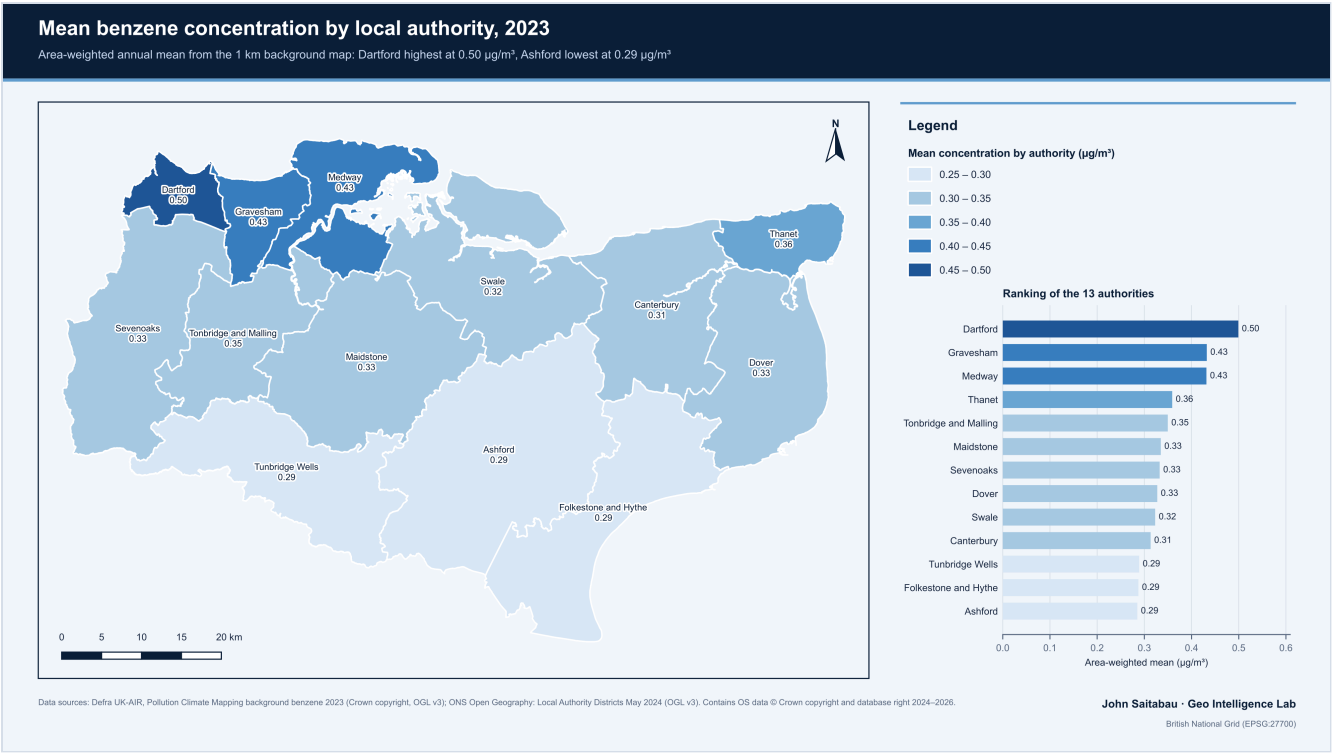
Method

1. **Build the surface.** The national 1 km PCM grid was converted to a raster and clipped to the 12 Kent districts and Medway. The PCM map was chosen over Defra's LAQM background maps because the LAQM benzene projections still rest on a 2001 base year.
2. **Summarise by authority.** The grid was resampled to 100 m by nearest neighbour, which leaves values unchanged. Each authority's mean therefore weights every 1 km cell by the area it covers inside the boundary, not by whether its centre falls inside.
3. **Allocate residents.** Each Output Area's 2021 usual residents took the value of the 1 km cell under its population-weighted centroid. Residents were then totalled by concentration class and authority.
4. **Test the emitters.** Each NAEI site was given the concentration of its own cell. Spearman's rank correlation measures whether larger emitters sit in higher-concentration cells.

Why not interpolate between emission sites?

Interpolating facility emissions (for example with IDW) produces a surface in tonnes per year spread across the land. That is not a concentration, and it ignores road traffic and the urban background, which dominate benzene in the UK. A modelled concentration surface answers the exposure question directly, and it can be compared with the emitters afterwards.

Finding 2 · A clear gradient from the estuary to the Weald

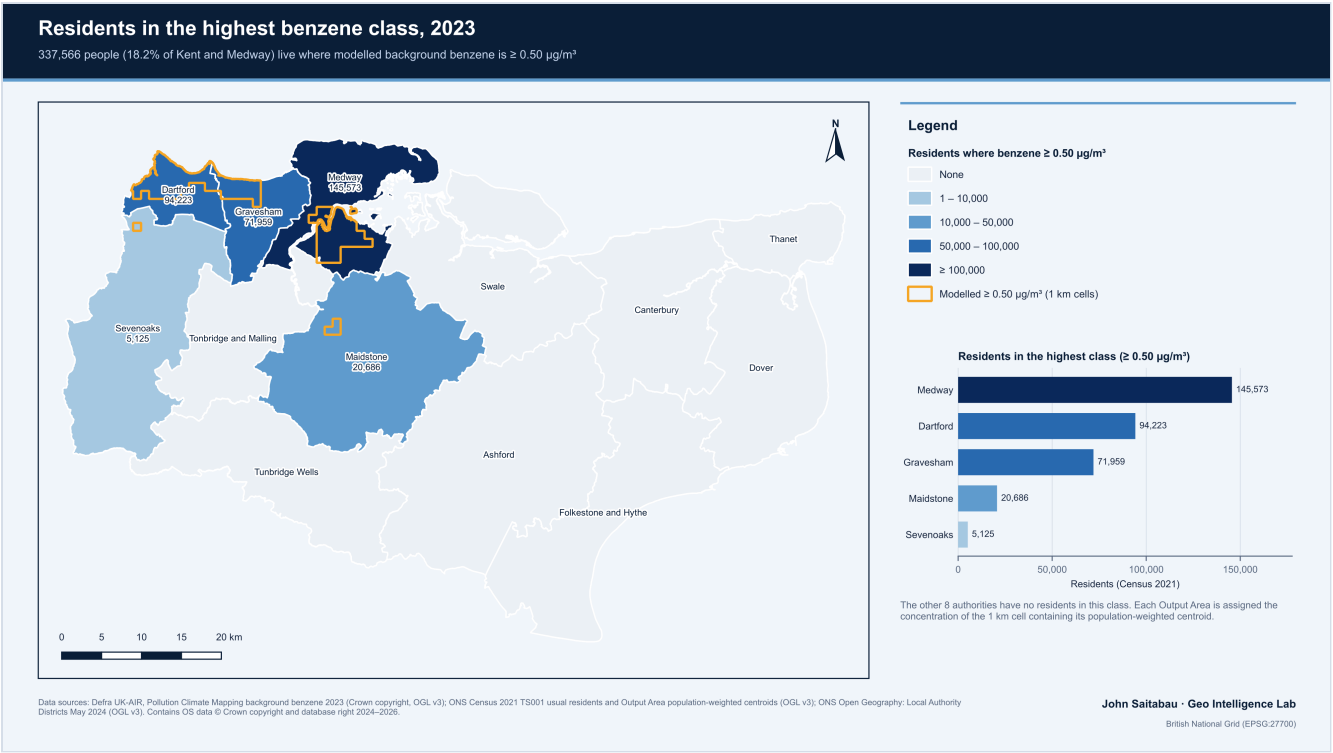


Area-weighted means fall from 0.499 µg/m³ in Dartford to 0.285 µg/m³ in Ashford. The three Thames-side authorities lead clearly. The rural south and east, from the Weald to Romney Marsh, sit at the bottom.

Authority	Area (km²)	Mean	Min	Max
Dartford	73	0.499	0.409	0.627
Medway	194	0.432	0.332	0.620
Gravesham	99	0.432	0.351	0.605
Thanet	104	0.359	0.300	0.482
Tonbridge and Malling	240	0.350	0.287	0.501
Maidstone	393	0.335	0.267	0.516
Sevenoaks	370	0.332	0.282	0.512
Dover	315	0.327	0.295	0.428
Swale	373	0.323	0.279	0.480
Canterbury	309	0.313	0.281	0.415
Tunbridge Wells	331	0.289	0.263	0.441
Folkestone and Hythe	357	0.287	0.254	0.427
Ashford	581	0.285	0.263	0.407

Concentrations in µg/m³. Means are area-weighted from the 1 km grid. Min and max are the extreme 1 km cells touching each authority.

Finding 3 · One resident in five lives in the top band



337,566 people (18.2%) live where modelled background benzene is at least 0.50 µg/m³. Only 5 of the 13 authorities have anyone in this band. Medway, Dartford and Gravesham account for 92% of them, and in Dartford it covers four residents in five.

Authority	Residents	≥ 0.50 µg/m³	Share	Pop.-weighted mean
Medway	279,769	145,573	52.0%	0.508
Dartford	116,752	94,223	80.7%	0.537
Gravesham	106,891	71,959	67.3%	0.520
Maidstone	175,793	20,686	11.8%	0.417
Sevenoaks	120,504	5,125	4.3%	0.385
Thanet	140,584	0	0.0%	0.411
Ashford	132,754	0	0.0%	0.338
Tonbridge and Malling	132,193	0	0.0%	0.393
Dover	116,403	0	0.0%	0.373
Swale	151,683	0	0.0%	0.395
Folkestone and Hythe	109,750	0	0.0%	0.347
Canterbury	157,427	0	0.0%	0.355
Tunbridge Wells	115,308	0	0.0%	0.351
Kent + Medway	**1,855,811**	**337,566**	**18.2%**	**0.416**

The count depends on where the top break sits. A cut-off of 0.48 µg/m³ gives 417,126 people and 0.52 µg/m³ gives 254,523, so treat it as a range rather than an exact figure.

Finding 4 · The biggest emitters are not where the air is worst

The 88 NAEI point sources in the study area reported 12.3 tonnes of benzene in 2023, and three sites account for 74% of it. The largest, a combined heat and power plant at Kemsley, sits in a cell of 0.40 µg/m³. That is below the Dartford average of 0.499 µg/m³. Across all sites, emission size and the concentration of the surrounding cell are almost unrelated (Spearman ρ = 0.11).

Site	Authority	Sector	t / year	Cell µg/m³
Kemsley	Swale	Paper, printing & publishing industries	7.37	0.396
Kemsley Mill CHP	Swale	Paper, printing & publishing industries	1.14	0.396
Isle of Grain LNG Installation	Medway	Processing & distribution of natural gas	0.58	0.362
Kemsley Generating Station	Swale	Waste collection, treatment & disposal	0.43	0.386
Allington Quarry	Tonbridge and Malling	Waste collection, treatment & disposal	0.39	0.424
Smurfitt Kappa Townsend Hook	Tonbridge and Malling	Paper, printing & publishing industries	0.34	0.474
Ashford	Ashford	Waste collection, treatment & disposal	0.21	0.325
Kingsnorth Oil Treatment Plant	Medway	Waste collection, treatment & disposal	0.20	0.418
Margate	Thanet	Chemical industry	0.20	0.402
Aylesford	Tonbridge and Malling	Waste collection, treatment & disposal	0.20	0.412

64 of the 88 sites, carrying 99.8% of the tonnage, are flagged in NAEI as wholly or partly modelled rather than reported by the operator. Site and operator names are as published by NAEI.

Limitations

- **Background only.** At 1 km the PCM map smooths out roadside and near-plant peaks, so kerbside exposure is higher than shown.
- **No local check.** No national-network benzene monitor was found in Kent or Medway. Local diffusion-tube data could not be retrieved while UK-AIR was offline in October 2026, so the surface is not validated against measurements.
- **Approximate allocation.** Residents are placed at Output Area centroids, which hold about 324 people each on average, not at home addresses.
- **Mixed years.** Concentrations and emissions are for 2023, and population is from Census 2021.
- **Annual means.** Short episodes, such as a plant upset or a winter inversion, do not show in a yearly average.

Who this helps

- **Council air-quality officers:** where a diffusion-tube survey would add most, starting with the Thames-side towns.
- **Public health teams:** which communities to prioritise for risk communication.
- **Planners:** background context for housing and school sites along the estuary corridor.
- **Residents:** a sourced, plain-language picture of the air where they live.

Need an analysis like this for your area?

I build air quality, exposure and site-screening studies from open and client data, from data sourcing to print-ready maps. See more work and get in touch at **johnsaita.github.io**.